EVALUATING A CLASSIFIER

The Confusion Matrix, and the metrics built from it

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MOTIVATION

A single number isn't enough

In our Default example, a classifier that **always predicts “No default”** is right **96.7% of the time**, simply because defaults are rare.

> *That model is useless: it never catches a single real default. A single accuracy number can hide exactly the failure we care about.*

We need to look at the full breakdown of correct and incorrect predictions (the **confusion matrix**) and derive several metrics from it.

Anatomy of a 2×2 confusion matrix

	Actual: Positive	Actual: Negative
Predicted: Positive	**TP** **True Positive** *correctly predicted positive*	**FP** **False Positive** *wrongly predicted positive*
Predicted: Negative	**FN** **False Negative** *wrongly predicted negative*	**TN** **True Negative** *correctly predicted negative*

Rows = what the model predicted.
Columns = what actually happened.
Every metric on the next slides is built from just these four counts.

Confusion Matrix & Classification Metrics

	Actual: Positive	Actual: Negative
Predicted: Positive	**TP** *True Positive*	**FP** *False Positive*
Predicted: Negative	**FN** *False Negative*	**TN** *True Negative*

Rows = predicted, columns = actual

Accuracy

`(TP+TN) / Total`

overall, how often right

Precision

`TP / (TP+FP)`

of predicted +, how many right

Recall (Sensitivity)

`TP / (TP+FN)`

of real +, how many caught

Specificity

`TN / (TN+FP)`

of real –, how many caught

F1-score

`2×P×R / (P+R)`

harmonic mean of P & R

WORKED EXAMPLE

Back to our Default example

LDA on the Default dataset, positive class = "default = Yes" (threshold 0.5):

	Actual: Yes	Actual: No
Predicted: Yes	TP = 80	FP = 22
Predicted: No	FN = 253	TN = 9,645

Total = 10,000 customers. We'll reuse these four numbers on every slide that follows.

METRIC 1 · ACCURACY

Accuracy: how often right?

Of all predictions, what fraction were correct?

$$\text{Accuracy} = (TP + TN) / \text{Total}$$

Plugging in our Default numbers:

RESULT

```
(80 + 9,645) / 10,000

= 0.97   (97%)
```

Looks excellent, but we already saw that predicting “No” for everyone gets 96.7%. Accuracy alone can't tell the two apart.

METRIC 2 · PRECISION

Precision: when we say “Yes”, how often right?

Of all the customers we flagged as likely to default, how many actually did?

$$\text{Precision} = TP / (TP + FP)$$

Plugging in our Default numbers:

RESULT

```
80 / (80 + 22)

= 0.78   (78%)
```

High precision means few false alarms: important when acting on a “Yes” is costly (e.g. denying someone credit).

METRIC 3 · RECALL

Recall: of the real “Yes”s, how many did we catch?

Of all customers who actually defaulted, how many did we correctly flag?

$$\text{Recall} = TP / (TP + FN)$$

Plugging in our Default numbers:

RESULT

```
80 / (80 + 253)

= 0.24   (24%)
```

Also called Sensitivity or the True Positive Rate. This is the weak point of our model: it misses 76% of real defaulters.

THE TRADE-OFF

Precision vs. Recall: a tug of war

Lowering the decision threshold catches more real positives (↑ recall) but also raises more false alarms (↓ precision) - you rarely improve both at once.

P

High Precision priority

Minimize false alarms.

Example: spam filter, don't want to lose real emails to the spam folder.

R

High Recall priority

Minimize missed positives.

Example: cancer screening, don't want to miss a real case, even with more false alarms.

METRIC 4 · SPECIFICITY

Specificity: recall's mirror image

Of all customers who did NOT default, how many did we correctly leave alone?

$$\text{Specificity} = TN / (TN + FP)$$

Plugging in our Default numbers:

RESULT

```
9,645 / (9,645 + 22)

= 0.998   (99.8%)
```

Also called the True Negative Rate. Our model is excellent at correctly recognizing non-defaulters - recall (for the "Yes" class) is its real weak spot.

METRIC 5 · F1-SCORE

F1-score: balancing precision & recall

A single number that summarizes both precision and recall together:

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$$

The harmonic mean: it stays low unless both precision AND recall are reasonably high.

Plugging in our Default numbers (precision = 0.78, recall = 0.24):

RESULT

$$2 \times (0.78 \times 0.24) / (0.78 + 0.24)$$

$$= 0.37$$

Notice F1 (0.37) is much closer to the lower of the two (recall = 0.24) than to their simple average (0.51).
That's the harmonic mean punishing the weak recall.

SUMMARY

All the metrics, side by side

Same confusion matrix (TP=80, FP=22, FN=253, TN=9,645) feeding every formula:

Metric	Formula	Value
Accuracy	(TP+TN) / Total	0.97
Precision	TP / (TP+FP)	0.78
Recall (Sensitivity)	TP / (TP+FN)	0.24
Specificity	TN / (TN+FP)	0.998
F1-score	2×P×R / (P+R)	0.37

One matrix, five different stories. **Which one matters most depends entirely on what mistake is more costly for your problem.**

PRACTICAL GUIDANCE

So, which metric should I use?

Classes roughly balanced, mistakes equally costly	**Accuracy**
False positives are costly (e.g. denying a good customer credit)	**Precision**
False negatives are costly (e.g. missing a fraud, a disease)	**Recall**
Need one balanced number, classes imbalanced	**F1-score**

KEY TAKEAWAYS

Wrapping up

- The confusion matrix (TP, FP, FN, TN) is the raw material behind every classification metric
- Accuracy can be misleading on imbalanced classes (it's not the whole story)
- Precision: of predicted positives, how many were right. Recall: of real positives, how many were caught
- Precision and recall trade off against each other as you move the decision threshold
- F1-score balances both into one number; specificity is recall's mirror for the negative class